Summary of: “HYPERBOLIC QUADRATURE METHOD OF MOMENTS FOR THE ONE-DIMENSIONAL KINETIC EQUATION” by Fox-Laurent-2022

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1 Framework and Definitions

The problem addressed concerns moment closures for the one-dimensional (1-D) kinetic equation describing the evolution of a velocity distribution function (VDF) $f(t, x, u)$. The objective is to construct a closure for the moment of order $2n + 1$ based on known realizable moments up to order $2n$, such that the resulting moment system is globally hyperbolic and in conservative form.

1.1 Kinetic Equation and Raw Moments

We consider the 1-D kinetic equation including only the free-transport term :

$$\partial_t f + u\partial_x f = 0 \tag{1}$$

The raw moments of order k of the VDF are defined by integration over the velocity space :

$$M_k(t, x) := \int_{\mathbb{R}} f(t, x, u)u^k \, du, \quad k \in \{0, 1, \dots, 2n + 1\} \tag{2}$$

The vector of known moments is denoted by $\mathbf{M}_{2n} = (M_0, M_1, \dots, M_{2n})^t$. The spatial flux for the moment M_k depends on the moment of order $k + 1$, leaving the system unclosed for the highest-order moment in the vector.

1.2 Existing Closure Methods

Historically, closing the moment system has been achieved through perturbative methods such as Grad’s 13-moment closure or non-perturbative reconstructions like Levermore’s entropy maximization. However, these approaches often lose global hyperbolicity or become computationally expensive, in contrast to quadrature-based moment methods (QMOM) which ensure realizability on the boundary of moment space.

1.3 HyQMOM in 2018

In its 2018 formulation *(see "Conditional Hyperbolic Quadrature Method of Moments for Kinetic Equations" by Fox, Laurent and Vié)*, the Hyperbolic Quadrature Method of Moments (HyQMOM) was limited to low-order systems, specifically moments up to the fourth order. By setting the recurrence coefficient $a_2 = 0$, one eigenvalue of the Jacobian matrix was forced to equal the mean velocity $\bar{u}$. While this "trick" ensured global hyperbolicity for low orders, it did not provide a straightforward path to extend the closure to higher-order moments ($n > 2$).

1.4 Centered and Standardized Moments

To facilitate the construction of the closure, centered and standardized moments are used to make the problem invariant to translation and scaling.

— **Centered Moments** : For $M_0 > 0$, these are defined relative to the mean velocity $\bar{u} = M_1/M_0$:

$$C_k := \frac{1}{M_0}\int_{\mathbb{R}} f(t,x,u)(u-\bar{u})^k \, du \tag{3}$$

where $C_0 = 1$, $C_1 = 0$, and C_2 represents the velocity variance.

— **Standardized Moments** : For $C_2 > 0$, these normalize the centered moments by the variance :

$$S_k := \frac{C_k}{C_2^{k/2}} \tag{4}$$

By definition, $(S_0, S_1, S_2) = (1, 0, 1)$.

Of course one can see that to go from moments to centered or standard moments, we apply respectively the change of variable :

1. $$v \to \frac{1}{M_0} f(\bar{u} + v)$$

2. $$v \to \frac{\sqrt{C_2}}{M_0} f(\bar{u} + \sqrt{C_2} v)$$

To go from S_k and C_k to M_k we will need : $\tilde{C_k} = (M_0, \bar{u}, C_2, C_3, \ldots, C_n)$ and $\tilde{S}_n = (M_0, \bar{u}, C_2, S_3, \ldots, S_n)$.

Finally let us define the functionals that will be useful later :

— $$\langle P \rangle_{M_n} = \int_{\mathbb{R}} P(u) f(u) du$$

— $$\langle X^k \rangle_{S_N} = S_k$$

We therefore have :

$$\langle P(X) \rangle_{M_N} = M_0 \langle P(\bar{u} + \sqrt{C_2} X) \rangle_{S_N} \tag{5}$$

2 Formulation of the Moment System

The governing equations for the 1-D kinetic model are formulated by taking the moments of the VDF. We define the moment vector of order N as $\mathbf{M}_N = (M_0, M_1, \ldots, M_N)^T$. The evolution of these moments is governed by the following system of conservation laws :

$$\partial_t \mathbf{M}_N + \partial_x \mathbf{F}(\mathbf{M}_N) = 0 \tag{6}$$

where $\mathbf{F}(\mathbf{M}_N) = (M_1, M_2, \ldots, M_N, M_{N+1})^T$ represents the flux vector. To close this system, the highest-order moment M_{N+1} must be expressed as an algebraic function of the known vector $\mathbf{M}_N$.

2.1 Rewriting the equation

Some mathematical properties (such as hyperbolicity) are better analyzed using central moments. By applying the chain rule to the conservative system, we obtain :

$$\partial_t \tilde{\mathbf{C}}_N + \mathbf{J}\partial_x \tilde{\mathbf{C}}_N = 0 \tag{7}$$

The matrix $\mathbf{J}$ is the Jacobian of the system in the central moment space, defined as :

$$\mathbf{J} = \left(\frac{\partial \mathbf{M}}{\partial \tilde{\mathbf{C}}}\right)^{-1} \frac{\partial \mathbf{F}}{\partial \mathbf{M}} \frac{\partial \mathbf{M}}{\partial \tilde{\mathbf{C}}} \tag{8}$$

2.2 Characteristic Polynomial Properties

The stability of the system depends on the roots of the characteristic polynomial of $\mathbf{J}$.

[Proposition 3.1] Let $\mathbf{M}_N$ be a realizable moment vector with $M_0 > 0$ and $C_2 > 0$. Let $\bar{P}_{N+1}(X)$ be the characteristic polynomial of $\mathbf{J}$. If the standardized closure S_{N+1} is independent of $(M_0, \bar{u}, C_2)$, then the scaled characteristic polynomial :

$$P_{N+1}(X) = \bar{P}_{N+1}(\bar{u} + C_2^{1/2})C_2^{-(N+1)/2} = |\mathbf{J} - (\bar{u} + C_2^{1/2}X)\mathbf{I}|C_2^{-(N+1)/2} \tag{9}$$

depends only on the standardized moments $(S_3, \ldots, S_N)$.

This prop's proof is a bit long but in a nutshell : we want the polynom of (5) but in (S_k). We can apply the change of variable seen previously :

$$v \to \frac{1}{M_0} f(\bar{u} + v)$$

This then gives equations from which we can find an expression for S_k.

[Theorem 3.2] The scaled characteristic polynomial $P_{N+1}(X) = \sum_{m=0}^{N+1} c_m X^m$ has coefficients c_m determined by the partial derivatives of the closure S_{N+1}. Specifically, for $m \in \{3, \dots, N\}$:

$$c_m = -\frac{\partial S_{N+1}}{\partial S_m}, \quad c_{N+1} = 1 \tag{10}$$

The remaining coefficients c_0, c_1, c_2 are linear combinations of S_m and c_m, ensuring the polynomial satisfies the internal conservation constraints. From the definition of c_0, c_1, c_2 we can show that :

$$\langle P_{N+1} \rangle = 0, \quad \langle P'_{N+1} \rangle = 0, \quad \langle X P'_{N+1} \rangle = 0$$

these equalities will be useful later...

This proof consists of saying that if P_{N+1} doesn't depend on $(M_0, \bar{u}, C_2)$, which is the case ; we can choose $(1, 0, 1)$ so that $S_N = C_N = M_N$. This tells us that P_{N+1} is equal to the polynom of $\frac{\partial F}{\partial M}$. Since we know the form of that Jacobi matrix, we can deduce the coefficient to be

$$-\frac{\partial M_{N+1}}{\partial M}$$

using the chain rule we deduce they are equal to

$$-\frac{\partial M_{N+1}}{\partial S} \left(\frac{\partial M}{\partial S}\right)^{-1}$$

3 The closure

We want to close the system. The system is closed and globally hyperbolic if the eigenvalues of P_{N+1} are both real and distinct. In HyQMOM, this is achieved by imposing a specific structure on the characteristic polynomial.

3.1 Theorem 4.6 and the Factorization

The fundamental result of the HyQMOM method is expressed in the following theorem :

[Theorem 4.6] For any strictly realizable standardized moment vector $\mathbf{S}_{2n}$, there exists a unique closure S_{2n+1} such that the standardized characteristic polynomial $P_{2n+1}(X)$ factorizes as :

$$P_{2n+1}(X) = Q_n(X) R_{n+1}(X) \tag{11}$$

where $Q_n(X)$ is the n-th order monic orthogonal polynomial. And the polynomial $R_{n+1}(X)$ is defined by the recurrence relation :

$$R_{n+1}(X) = (X - \alpha_n) Q_n(X) - \beta_n Q_{n-1}(X) \tag{12}$$

if and only if the final coefficients α_n and β_n are defined as :

$$\alpha_n = a_n = \frac{1}{n} \sum_{k=0}^{n-1} a_k, \quad \beta_n = \frac{2n+1}{n} b_n$$

3.2 proof's idea

This proof is done in 2 parts. The first part is to say that if

$$P_{2n+1}(X) = Q_n(X)R_{n+1}(X)$$

Then conditions :

$$\langle P_{N+1}\rangle = 0, \quad \langle P'_{N+1}\rangle = 0, \quad \langle XP'_{N+1}\rangle = 0$$

are equivalent to saying that :

$$\alpha_n = a_n = \frac{1}{n}\sum_{k=0}^{n-1} a_k, \quad \beta_n = \frac{2n+1}{n} b_n$$

This is more or less immediate by developing the polynoms using as defined in orthogonal polynomials :

$$a_k = \frac{\langle XQ_k^2\rangle}{\langle Q_k^2\rangle}, \quad b_k = \frac{\langle Q_k^2\rangle}{\langle Q_{k-1}^2\rangle}$$

The Second part is to prove that :

$$P_{2n+1}(X) = Q_n(X)R_{n+1}(X)$$

This is a bit more complex and done on Matlab. It goes like this : Given $(a_k)_{k\in(1,\dots,n-1)}$ and $(b_k)_{k\in(2,\dots,n)}$, compute a_n then use reverse Chesbychev algorithm to find $S_k = \sigma_{0,k}$. We then create a vector $Y = (a_1, b_2, a_2, \dots, a_{n-1}, b_n)$ and use it to find c_k with the relation found in theorem 3.2. Then subtract P_{2n+1} and Q_nR_{n+1}

3.3 Hyperbolicity

As said previously, the method proposed is globally hyperbolic.Short explanation of this is because the roots of Q_n are all distinct and reel, and the roots of R_{n+1} are reel, distinct as well and separate those of Q_n). The first result is proved in another research paper *(see 2004 "orthogonal polynomials" by W. Gautschi)* and the second one comes from Christoffel– Darboux formula, implying that : $R_{n+1}(x_k)R_{n+1}(x_{k+1}) < 0$ where $(x_k)_{k\in\{1,\dots,n\}}$ are the roots of Q_n

4 Simplification of the method

We have seen a way of closing the equation using the standardized moments. We will now, in this part prove that we can actually only focus on M_n and not bother computing S_n.

In a few words : the idea is to prove that

$$P_{2n+1}(X) = Q_n(X)R_{n+1}(X) \quad \textit{is equivalent to} \quad \bar{P}_{2n+1}(X) = \bar{Q}_n(X)\bar{R}_{n+1}(X)$$

First of all P is linked to $\bar{P}$ with relation (9) Let Q_k be the monic orthogonal polynomial for $\langle.\rangle_{S_k}$. Then the monic orthogonal polynomial, $\bar{Q}_k$ for $\langle.\rangle_{M_k}$ is defined by :

$$\bar{Q}_k(X) = \sqrt{C_2}^k Q_k\left(\frac{X-\bar{u}}{\sqrt{C_2}}\right) \tag{13}$$

this is proven by checking the definition with (5)
We can then apply the change of variable :

$$\bar{a}_n = \frac{\langle X\bar{Q}_n^2\rangle_{M_{2n}}}{\langle \bar{Q}_n^2\rangle_{M_{2n}}} = \frac{\langle(\bar{u}+\sqrt{C_2}X)Q_n^2\rangle}{\langle Q_n^2\rangle} = \bar{u}+\sqrt{C_2}a_n \tag{14}$$

$$\bar{b}_n = \frac{\langle \bar{Q}_n^2\rangle_{M_{2n}}}{\langle \bar{Q}_{n-1}^2\rangle_{M_{2n}}} = C_2\frac{\langle Q_n^2\rangle}{\langle Q_{n-1}^2\rangle} = C_2 b_n \tag{15}$$

This gives the relation :

$$\bar{R}_{n+1}(X) = \sqrt{C_2}^{n+1} R_{n+1}\left(\frac{X-\bar{u}}{\sqrt{C_2}}\right) \tag{16}$$

We then use (9),(13),(16) to show the equivalence.

5 The algorithm

Algorithm 1: Computation of M_{2n+1}.

Data: M_{2n} strictly realizable
Result: M_{2n+1}
Initialize each scalar $\sigma_{k,p}$ to zero for $k \in \{-1,\dots,n\}$, $p \in \{0,\dots,2n+1\}$;
for $p \leftarrow 0$ **to** $2n$ **do**
  $\sigma_{0,p} \leftarrow M_p$;
$(\bar{a}_0, \bar{b}_0) \leftarrow (\frac{M_1}{M_0}, 0)$;
for $k \leftarrow 1$ **to** $n-1$ **do**
  for $p \leftarrow k$ **to** $2n-k$ **do**
    $\sigma_{k,p} \leftarrow \sigma_{k-1,p+1} - \bar{a}_{k-1}\sigma_{k-1,p} - \bar{b}_{k-1}\sigma_{k-2,p}$;
  $\bar{a}_k \leftarrow \frac{\sigma_{k,k+1}}{\sigma_{k,k}} - \frac{\sigma_{k-1,k}}{\sigma_{k-1,k-1}}$;
  $\bar{b}_k \leftarrow \frac{\sigma_{k,k}}{\sigma_{k-1,k-1}}$;
$\sigma_{n,n} \leftarrow \sigma_{n-1,n+1} - \bar{a}_{n-1}\sigma_{n-1,n} - \bar{b}_{n-1}\sigma_{n-2,n}$;
$\bar{b}_n \leftarrow \frac{\sigma_{n,n}}{\sigma_{n-1,n-1}}$;
$\bar{a}_n \leftarrow \frac{1}{n}\sum_{k=0}^{n-1}\bar{a}_k$; `// Set the closure`
$\sigma_{n,n+1} \leftarrow \sigma_{n,n}(\bar{a}_n + \frac{\sigma_{n-1,n}}{\sigma_{n-1,n-1}})$; `// Reverse Chebyshev algorithm`
for $k \leftarrow n-1$ 0 **do**
  $\sigma_{k,2n-k+1} \leftarrow \sigma_{k+1,2n-k} + \bar{a}_k\sigma_{k,2n-k} + \bar{b}_k\sigma_{k-1,2n-k}$;
$M_{2n+1} \leftarrow \sigma_{0,2n+1}$;